

LAYOUT PLANING MODELS AND DESIGN ALGORITHMS

CH #6

6.1 Introduction

- A facilities layout strategy should emerge from the overall strategic plan.
- Product, manufacturing, marketing distribution, management, and human resource plans will be impacted by and will impact on the facilities layout.
- It is important question which come first. If material handling is consider first it will effect the following decisions.
 1. Centralized versus decentralized storage of work-in-process (WIP), tooling, and, supplies.
 2. Fixed path versus variable path handling.
 3. The handling unit (unit load) planned for the systems.
 4. The degree of automation used in handling.
 5. The type of inventory control, physical control, and computer control of materials.

6.2 Basic Layout Types

1. Fixed Material Location Departments
2. Production Line Departments
3. Product Family Departments
4. Process Departments

6.3 Layout Procedures

In designing layouts, the procedures can be classified into two main categories:

1. Construction type layout methods basically involve developing a new layout “from scratch”, and
2. Improvement procedures generate layout alternatives based on an existing layout.

6.3.1 Apple's Plant Layout Procedure (Not necessary in that sequence)

1. Procure the basic data
2. Analyze the basic data
3. Design the productive process
4. Plan the material flow pattern
5. Consider the general material handling plan
6. Calculate equipment requirements
7. Plan individual workstations
8. Select specific material handling equipment
9. Coordinate groups of related operations
10. Design activity interrelationships
11. Determine storage requirements
12. Plan service and auxiliary activities
13. Determine space requirements
14. Allocate activities to total space
15. Consider building types
16. Construct master layout
17. Evaluate, adjust, and check the layout with the appropriate persons
18. Obtain approvals
19. Install the layout
20. Follow up on implementation of the layout

6.3.2 Reed's Plant Layout Procedure

1. Analyze the product or products to be produced
 2. Determine the process required to manufacture the product
 3. Prepare layout planning charts
 4. Determine workstations
 5. Analyze storage area requirements
 6. Establish minimum aisle widths
 7. Establish office requirements
 8. Consider personnel facilities and services
 9. Survey plant services
 10. Provide for future expansion
- Reed calls the layout planning chart "the most important single phase of the entire layout process"

LAYOUT PLANNING CHART

PART NO.1PART NAMEPLASTIC CONTACT PADPCS/ASSY1PCS REQ/HR70.4SHEET 1 OF 1

ASSY NO.—ASSY NAME—ASSY/PRODUCT2PRODUCTION HRS/DAY6.0PREPARED BYJ.G.D.DATE1-4-60

MATERIALPLASTICSIZE1 1/2" O.D. x 3/8" I.D. (from 4" x 8" x 1/4" SHEETS)PCS/DAY422LOT SIZE1APPROVED BYDATE

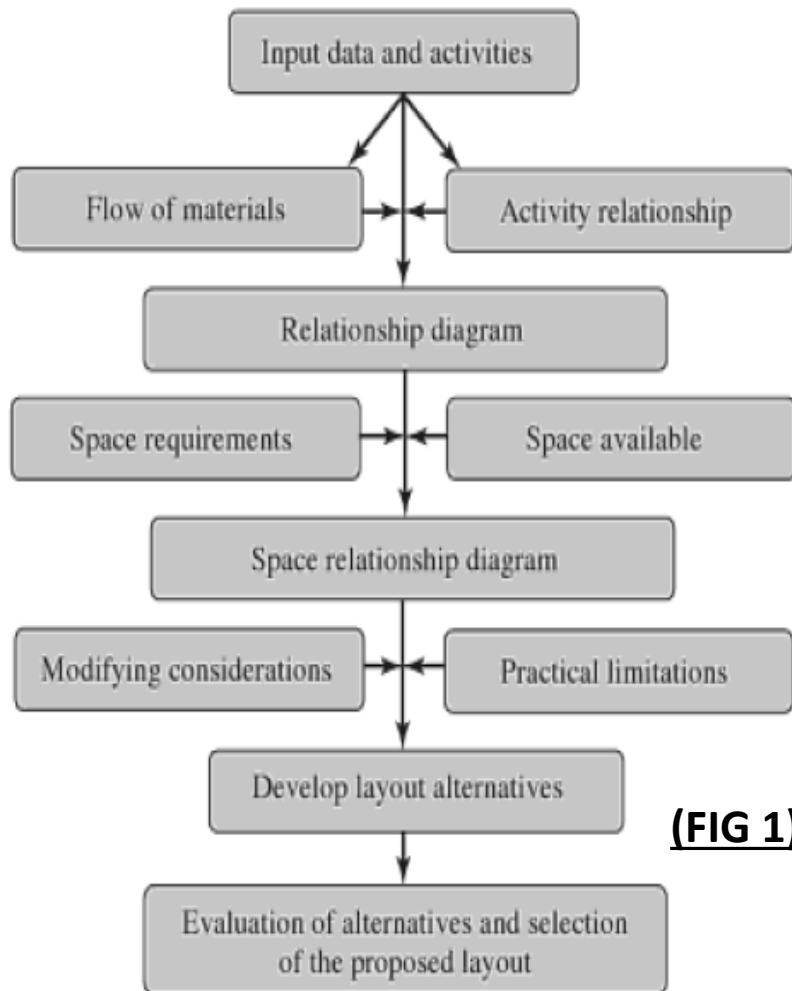
ST NO.	F M S I	DESCRIPTION	OPER NO.	DEPT NO.	TIME PER PIECE	MACHINE OR EQUIPMENT	MACHINES REQD			OPER PER MACH	TOTAL MANPOWER				HANDLING REQUIREMENTS				REMARKS
							MACH FRAC	COMB WITH	MACH REQD		CREQ FRAC	MAN FRAC	COMB WITH	MEN REQD	HOW MOVED	CONT TYPE	LOAD SIZE	DIST MOVED	
1	○●●●	FROM MATERIALS STORAGE													FREE WGT	PALLETS	4 SHEETS	100'	
2	○●●●	ON PALLET BY SAW		2															
3	○●●●	TO SAW TABLE																	
4	○●●●	SAW INTO STRIPS 2 1/2" x 8"	10	2	.02	TABLE SAW	.028	9-10 16-10	1	2	.028	.056	9-10 16-10	2					
5	○●●●	TO RACK BY SAW											1-20 14-30						
6	○●●●	IN RACK		2									17-20 3-20						
7	○●●●	TO HEATER											6-30 23-20						
8	○●●●	IN RACK BY HEATER		2									14-20						
9	○●●●	FEED INTO HEATER																	
10	○●●●	HEAT	10	2	.04	HEATER	.055	9-20 14-20	1	1	.055	.055	6-20 6-10	1					
	○●●●												2-10 2-10						
	○●●●												3-20 16-20						
	○●●●												13-10 14-10						
	○●●●												17-10						
11	○●●●	FEED TO PUNCH PRESS																	
12	○●●●	PUNCH TO SHAPE	10	2	.04	PUNCH PRESS	.055	4-10 4-20	1	1	.055	.055	4-10 4-20	1					
13	○●●●	TP BIN BY PUNCH PRESS						9-20 16-30					9-20 16-30						
14	○●●●	IN BIN		2				13A-10 19B-10					13A-10 19B-10						
15	○●●●	TO PARTS STORAGE											13A-20 5A-10		WHEEL AND TRUCK	TOTE BOX	400 PPR BOX	150'	
16	○●●●	IN PARTS STORAGE		5															
17	○●●●	TO ASSEMBLY													WHEEL AND TRUCK	TOTE BOX	400 PPR BOX	60'	
18	○●●●	IN BIN IN ASSEMBLY		5															
19	○●●●	TO TABLE																	
	○●●●																		
	○●●●																		

Reeds layout planning chart must incorporate the following objectives/ points

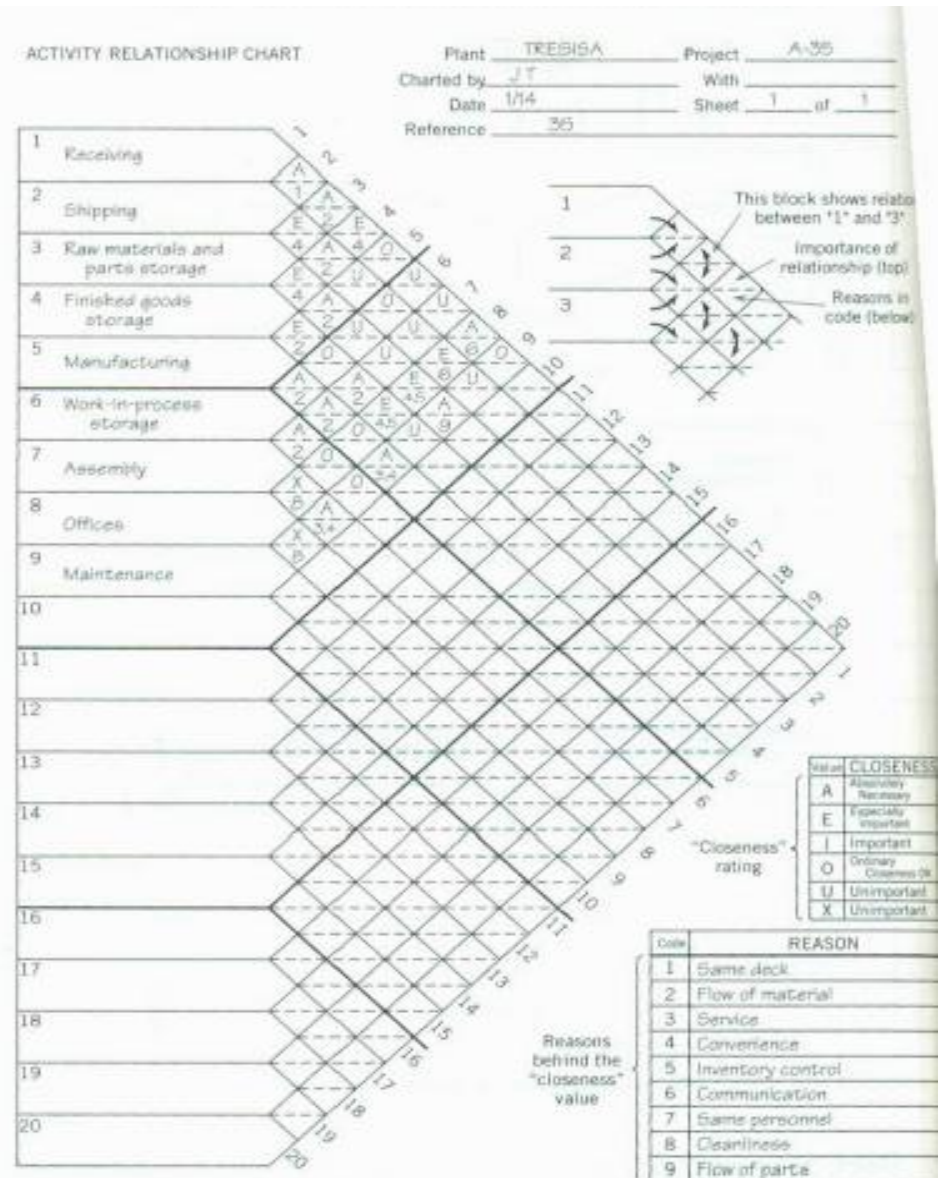
1. Flow process, including operations, transportation, storage, and inspection
2. Standard times for each operation
3. Machine selection and balance
4. Manpower selection and balance
5. Material handling requirement

6.3.3 Muther's Systematic Layout Planning (SLP) Procedure

- The systematic layout planning (SLP) procedure uses as its foundation the activity relationship chart (**FIG 2**)



(FIG 1)



(FIG 2)

- Based on the input data and an understanding of the roles and relationships between activities, a material flow analysis (from-to-chart) and an activity relationship analysis (activity relationship chart) are performed. From the analyses performed, a relationship diagram is developed. **(FIG 6.5)**

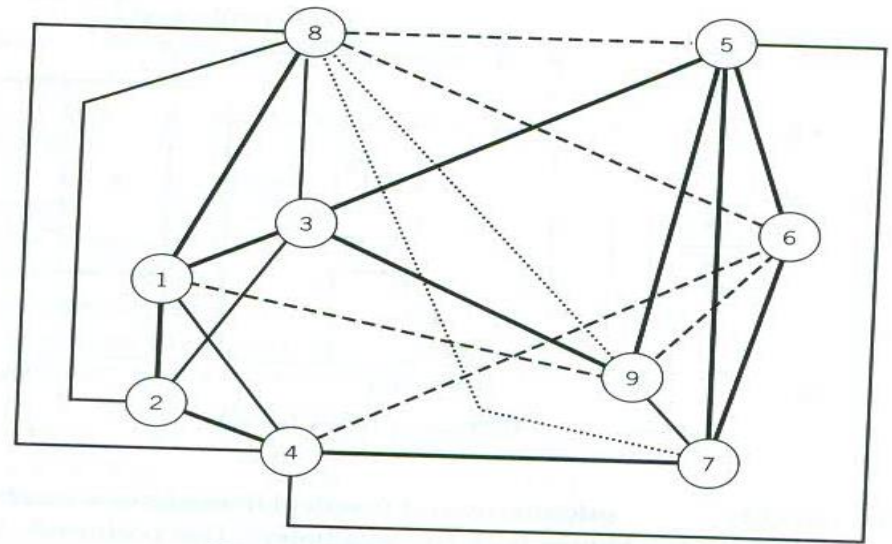


Figure 6.5 Relationship diagram.

- The next two steps involve the determination of the amount of space to be assigned to each activity. Departmental service and area requirement sheets would be completed for each department. Once the space assignments have been made, space templates are developed for each planning department and the space is “hung on the relationship diagram” to obtain the space relationship diagram **(FIG 6.6)**

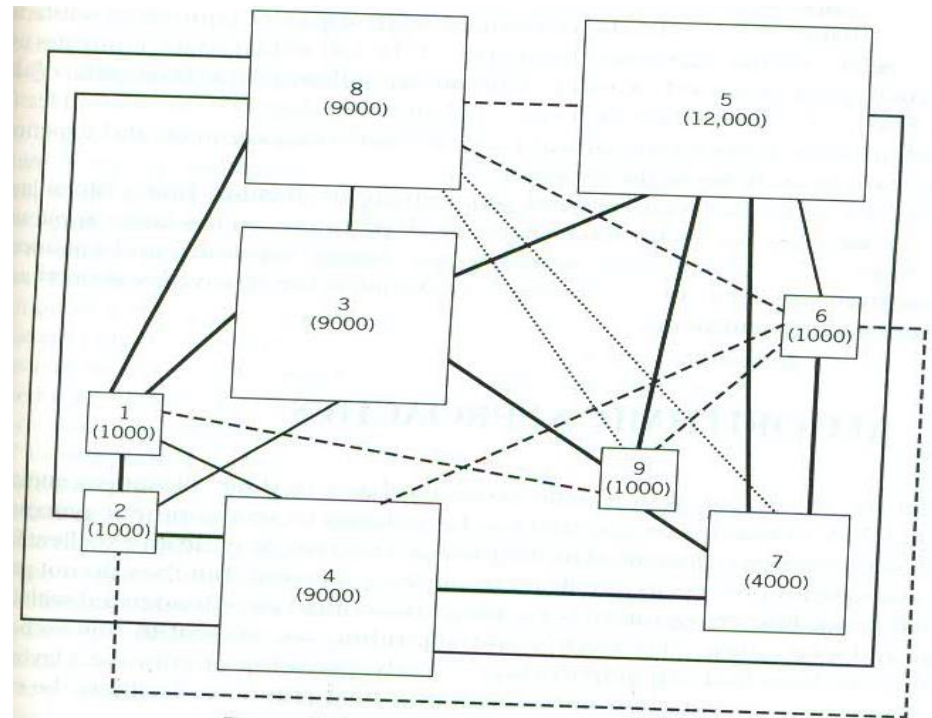
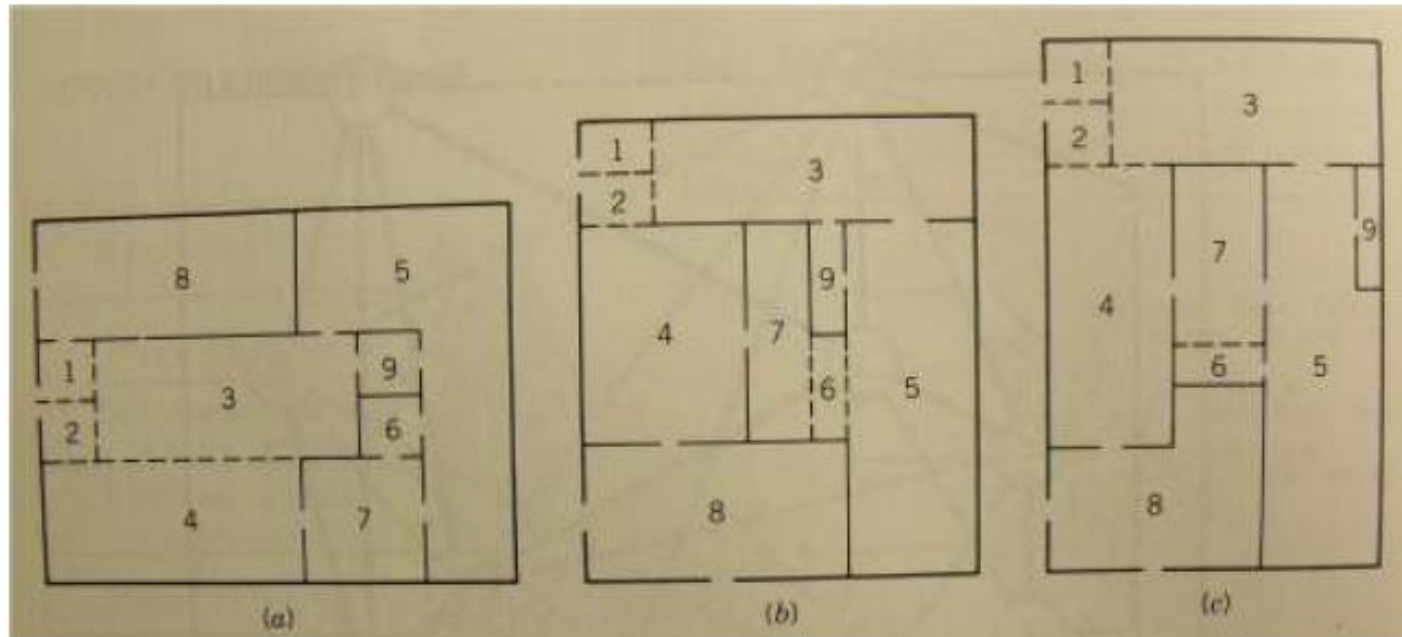


Figure 6.6 Space relationship diagram.

- Based on modifying considerations and practical limitations, a number of layout alternatives are developed and evaluated. The preferred alternative is then recommended.



6.4. Algorithmic Approaches

- In method discussed before we had relative placement of departments on the bases of their “closeness rating” or “material flow intensities” can only reduced algorithm processes but they don’t provide formal algorithm.

6.4.1 Algorithm classification

- Algorithm is classified according to type of input data they required.
 1. Qualitative “flow” data (such as relationship chart)
 2. Quantitative flow matrix expressed as a from-to charts.
 3. Qualitative and Quantitative both data (such as BLOCPLAN, **Factory Design Suite etc**)